

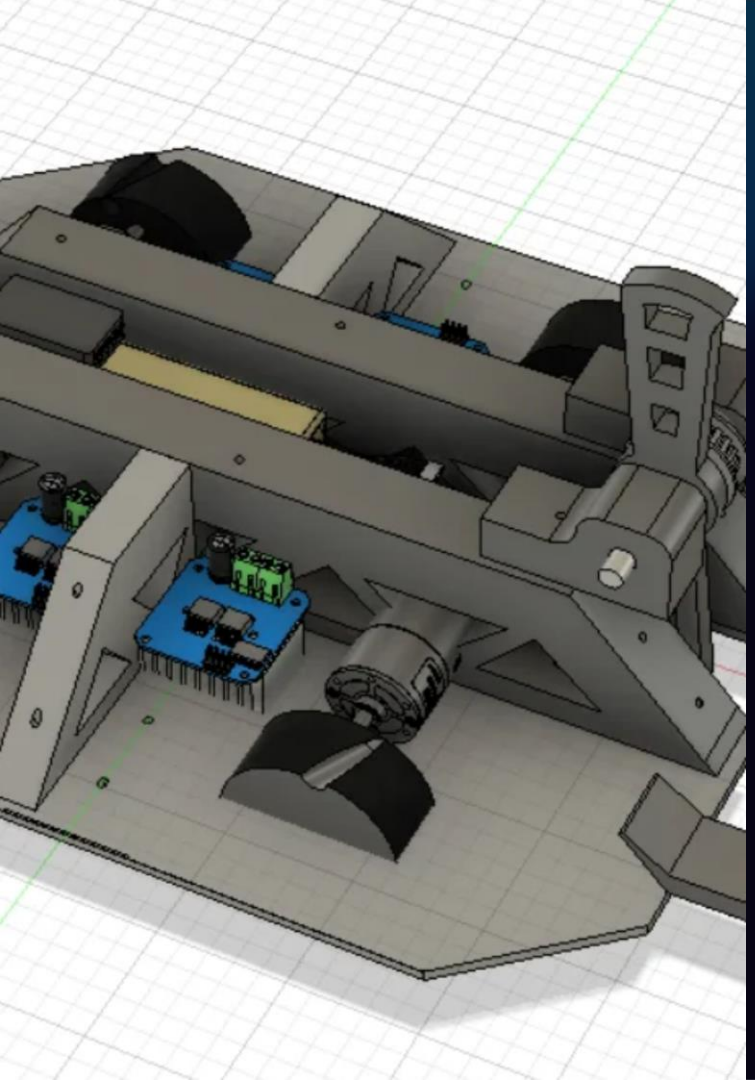
ROBOVITAP

— Build. Battle. Innovate.

A student's guide to entering VIT-AP's combat robotics championship

Register now

Rule Book



What is RoboWars?

RoboWars is not a coding contest. It is **real engineering** — mechanical design, electronics, and strategy colliding in a controlled combat arena. You build a wireless robot, you fight, and you win by making your opponent stop moving.

Format

team double-elimination bracket per weight category

Match Duration

3 minutes of active fight time per bout

Team Size

Up to 6 members, open to any institute background

COMPETE. WIN. EARN.

Prize Distribution — Win Big!

₹2,00,000 Total Prize Pool

Big rewards. Bigger bragging rights.

Beetleweight (1.36 kg)

₹30,000 Winner

₹15,000 Runner-up

₹5,000 2nd Runner-up

Category Total: ₹50,000

Dogeweight (8 kg)

₹40,000 Winner

₹20,000 Runner-up

₹10,000 2nd Runner-up

Category Total: ₹70,000

Featherweight (15 kg)

₹45,000 Winner

₹25,000 Runner-up

₹10,000 2nd Runner-up

Category Total: ₹80,000

Why Take Part?

What You Build

This is a portfolio-worthy project that combines hands-on mechanical fabrication, wireless electronics integration, and real team coordination under pressure. No other undergraduate activity gives you all three simultaneously.

Mechanical Design & Fabrication

Chassis, armor, weapon systems — built by your hands

Wireless Electronics

ESCs, LiPo batteries, receivers, kill switches — real embedded systems

Strategy Under Pressure

3-minute bouts demand split-second decisions and pre-match planning

What You Earn

Prize Pool

Tiered across weight categories of total prize pool of Rs. 2 Lakhs

Certificates

Official recognition for all participating teams

Arena Spotlight

Compete in front of ~20–30 teams from across institutes

Portfolio Project

A documented, real-world engineering build that stands out in any resume or interview

Robot Design Ideas: Horizontal / Vertical Spinner

High Damage Potential

High damage potential with rotating blades or discs.

Safety Braking

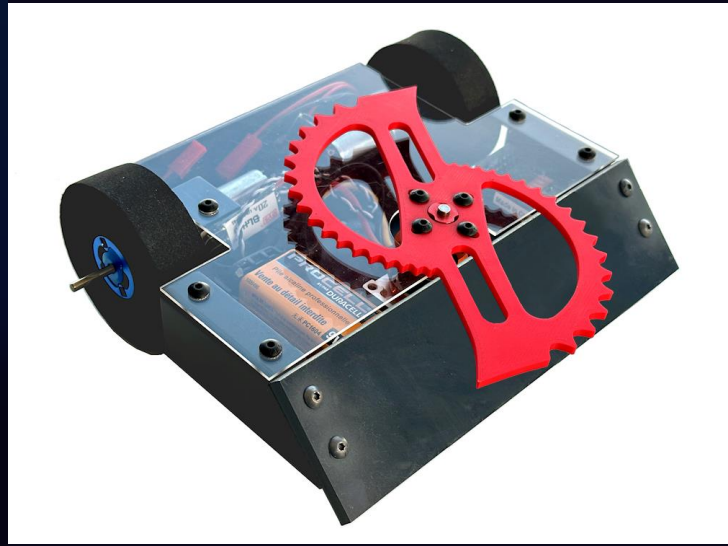
Must brake to a stop within 60 seconds of power-off.

Testing Restrictions

Can only be tested inside the arena.

Design Considerations

Requires careful weapon balancing and robust armor to survive counter-attacks.



Lifter / Grabber

Control-focused design that lifts or grabs opponents. Note: pinning/lifting alone cannot win a match — max 10 second hold before you must release. Effective for immobilizing aggressive spinners and controlling match pace.



Robot Design Ideas: Full-Body Spinner

Advanced design where the entire robot spins as a weapon. High build difficulty and high risk/reward. Requires expert-level engineering, perfect weight distribution, and exceptional armor. Only for experienced teams with significant budget.



Robot Design Ideas: Flipper (Spring/Pneumatic)

Launches opponents into the air or flips them over. Spring must stay locked outside the arena per rules — only unlocked when ready to fight. High crowd appeal and effective against unprepared opponents.



Robot Design Ideas: Wedge / Pusher

Simplest build, low risk, great first robot. Wins on control & positioning. A wedge-shaped front deflects opponent attacks while you push them around the arena. Perfect for beginners learning robot dynamics.



Choose Your Weight Category

Three categories are available. Your choice determines your budget, build complexity, and competitive landscape. Pick the one that matches your team's experience and resources — there is no shame in starting small.

Category	Weight Limit	Tolerance	Good For
Beetleweight	1.36 kg (3 lb)	1%	First-time builders, lower budget, faster build cycle
Dogeweight	8 kg	1%	Intermediate teams, room for a real weapon system
Featherweight	15 kg	1%	Experienced/well-funded teams, most competitive category

⚠ Weight is checked before AND after every match. All batteries must be on-board (remote controller weight excluded). A kill switch is compulsory in every category — no exceptions.

Estimated Build Cost: Beetleweight (1.36 kg)

Total Estimate

₹8,000 – ₹20,000

Cheapest category to enter — best for a first build. The wide range reflects whether your team shares a transmitter or buys its own, and whether you add a weapon system.

☆ RECOMMENDED FOR BEGINNERS

Component	Approx. Cost (₹)
Chassis (HDPE / 3D-printed PETG / thin aluminium)	1,500 – 3,000
Drive motors (2× micro geared DC/BLDC)	1,000 – 2,500
ESCs (2×, small brushed/brushless)	1,500 – 3,000
Battery (2S–3S LiPo, 450–850 mAh)	800 – 1,500
Receiver + Transmitter (shared/club Tx or entry-level own set)	0 – 6,400
Wheels, hardware, wiring, kill switch	1,500 – 3,000
Optional simple weapon (wedge/mini-spinner)	2,000 – 5,000

Estimated Build Cost: Dogeweight (8 kg)

Total Estimate

₹35,000 – ₹60,000

Intermediate category with room for a real weapon system. Motor and transmitter choices drive most of the cost variance. Budget with contingency — weapon fabrication often runs over.

INTERMEDIATE LEVEL

Component	Approx. Cost (₹)
Chassis (Aluminium / HDPE / polycarbonate)	3,000 – 6,000
Drive motors (2× — e.g. BBOX 15 w/ Mabuchi 775 or Nema17 Planetary BLDC)	7,400 – 12,000
ESCs (2×, rated to motor stall current)	4,000 – 8,000
Battery (3S–4S LiPo, 3–5 Ah)	2,500 – 5,000
Receiver + Transmitter (FlySky FS-i6 or Radiomaster)	6,400 – 18,000
Wheels, bearings, wiring, kill switch	3,000 – 6,000
Weapon system (motor + ESC + blade/lifter mechanism)	8,000 – 15,000

Estimated Build Cost: Featherweight (15 kg)

Total Estimate

₹50,000 – ₹90,000+

The most competitive and most expensive category. Aluminium 6061 chassis, high-current BLDC drive, and a serious spinner or flipper weapon are expected at this level. Teams should budget with meaningful contingency.

EXPERIENCED TEAMS ONLY

Component	Approx. Cost (₹)
Chassis (Aluminium 6061 / thick HDPE)	6,000 – 12,000
Drive motors (2× — e.g. BBOX 15 V2 w/ BLDC Saturn 3650)	12,000 – 16,000
ESCs (2×, higher current rated)	6,000 – 10,000
Battery (4S LiPo, 5–8 Ah)	5,000 – 8,000
Receiver + Transmitter (Radiomaster TX12 MKII recommended)	18,000 – 26,000
Weapon system (spinner/flipper: motor, ESC, blade, armor)	15,000 – 30,000
Armor, hardware, wiring, kill switch	6,000 – 10,000

All figures are approximate estimates based on current Indian robotics-component market pricing. Actual cost depends on design complexity, weapon type, and sourcing. Teams should budget with contingency.

Recommended Components: Drive Motors

Motor selection is one of the most consequential decisions in your build. The table below covers the four recommended options across all weight categories, with real specs and current market pricing.

Motor	Voltage	Torque	Speed	Weight	Price	Best For
Pro-Range Planetary Gear DC Motor PG36M555-19.2K	12V DC	45 N-cm	262 RPM	500 g	**₹1,800**	Beetleweight
Nema17 Planetary Geared BLDC Motor	24V	0.412 N-m	1080 RPM	650 g	**₹3,700**	Dogeweight
BBOX 15 with Mabuchi 775 DC Brushed Motor	18V DC	1.21 N-m	1200 RPM	832 g	**₹5,000**	Dogeweight / Featherweight
BBOX 15 V2 with BBOX Saturn 3650 Inrunner BLDC Motor	24V	17.82 N-m	1450 RPM	634 g	**₹6,000**	Featherweight (high torque)

The BBOX 15 V2 / Saturn 3650 combination delivers 17.82 N-m of torque — nearly 40× the Beetleweight option — making it the clear choice for Featherweight teams prioritizing pushing power and weapon drive.

Safety Rules Every Team Must Know

RoboWars is engineered for safety as much as competition. These are non-negotiable requirements — failure to comply means disqualification before you ever enter the arena.

Mandatory Hardware

1 Kill Switch

Compulsory in every category. Must be accessible and clearly marked. Judges will test it before every match.

2 On-Board Batteries Only

All power must be carried on the robot. Remote controller battery is the only exception to the weigh-in.

3 Tolerance on Weight

1% tolerance in all categories. Weighed before AND after every match. No grace margin exists.

Judging Criteria

Aggression

Active pursuit and engagement of the opponent throughout the bout

🎮 Control

Precise, deliberate movement — not just spinning in place

Damage

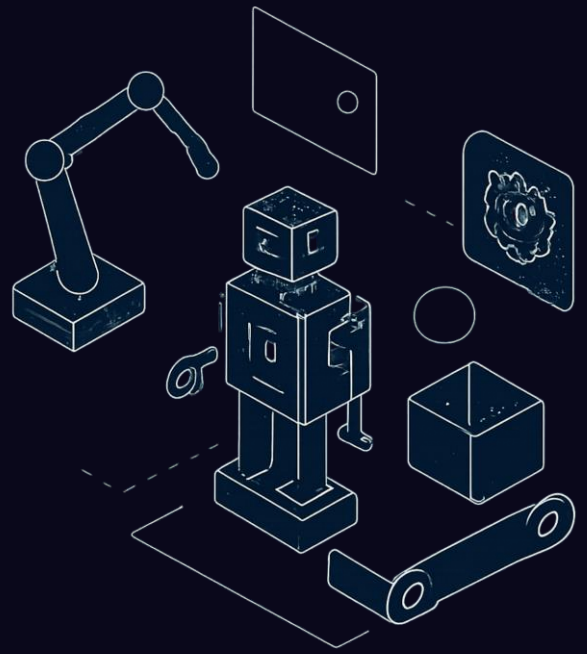
Visible, functional damage inflicted on the opponent's robot

Strategy

Tactical positioning, target selection, and match awareness

Ready to Build?

1. **Pick your category**
   (Beetleweight / Dogeweight / Featherweight)
2. **Budget using the estimates above**
3. **Fill the registration Google Form**
4. **Questions? Reach out to the RoboVITAP organizing team**



Rules Every Builder Must Know

RoboWars is engineered for safety as much as competition. These are non-negotiable requirements — failure to comply means disqualification before you ever enter the arena.

General & Safety Rules

- Wireless remote control only — all power on-board, kill switch (E-stop) compulsory
- Weight checked before and after every match — 0% tolerance
- Weapon lock required outside the arena; only removed once the bot is arena-ready

Competition Rules & Banned Items

Banned: projectiles, RF-jamming/electrical weapons, flying mechanisms, suction-cup anchoring

Spinning weapons must self-brake within 60 seconds of power cut, and can only be tested inside the arena

Matches are 3 minutes of active fight time; bots must engage at least once every 20 seconds — no tapping out except when immobile or time runs out

Victory = opponent immobilized; scoring on Aggression, Control, Damage, Strategy

Judges' decision is final at all times

Team & Registration Basics

- Team size: up to 6 members, from same institutes
- Unique team name + robot name (robot name can't be changed once set)
- Must submit a bot abstract; bots are inspected against that abstract
- Safety inspection happens the night before + re-verified before each match

YouTube link for complete info on various types of Battlebots - How to make Battlebots

Recommended Components: Batteries & Power Systems

The power system is the heart of your robot, dictating run time, burst power, and overall performance. Selecting the right battery and managing its discharge is crucial for sustained aggression and control during a match.

Battery Type	Voltage / Capacity	Max Discharge (C-rating)	Approx. Weight	Price	Best For
2S LiPo Pack	7.4V, 800-1300 mAh	45-75C	50-90 g	**₹800 – ₹1,500**	Beetleweight (Drive & Weapon)
3S LiPo Pack	11.1V, 2200-3000 mAh	60-100C	180-250 g	**₹2,500 – ₹4,000**	Dogeweight (Drive & Weapon)
4S LiPo Pack (Drive)	14.8V, 4000-6000 mAh	80-120C	350-500 g	**₹5,000 – ₹8,000**	Featherweight (Drive)
4S LiPo Pack (Weapon)	14.8V, 1500-2200 mAh	100-150C	130-180 g	**₹3,000 – ₹5,000**	Featherweight (High-Burst Weapon)

Always prioritize the battery's C-rating for weapons, especially spinners, as they demand immense current bursts. A higher C-rating means more power delivery without excessive voltage sag or risk of thermal runaway.

Recommended Components: Chassis Materials & Armor

The choice of chassis material directly impacts your robot's durability, weight, and fabrication difficulty. Select materials based on your weight class, budget, and weapon type.

Material	Key Properties	Pros	Cons	Best For
HDPE	Lightweight, impact-absorbing	Easy to machine, cost-effective, good for kinetic energy absorption	Low stiffness, vulnerable to cutting/abrasion	Beetleweight, light Dogeweight (chassis, wedge armor)
Aluminium 6061	Strong, lightweight, good stiffness	Excellent strength-to-weight, machinable, structural integrity	Can bend/dent from heavy impacts, more expensive than plastics	Dogeweight (chassis/frame), Featherweight (chassis, weapon components)
Hardox / AR500 Steel	Extremely hard, wear-resistant	Superior protection against spinners, high durability	Very heavy, difficult to machine, expensive	Featherweight (critical armor, weapon teeth/edges)
Titanium (Grade 5)	High strength-to-weight, corrosion-resistant	Best strength and resilience with minimal weight penalty	Very expensive, extremely difficult to machine	Featherweight (premium armor, critical stressed components)

Proper armor design is as crucial as material selection. Consider modular armor plates for easy replacement and effective angling to deflect opponent attacks rather than absorb them directly.

Contact Us

Questions? Get in Touch with the RoboVITAP Team.



Organizing Team

For general inquiries and support -

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Don’t wait. Register now and start you journey to the arena.

We're here to help you every step of the way. Don't hesitate to reach out!